

Paul He

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Education

University of Toronto

September 2021 - May 2025

Honours Bachelor of Science in Computer Science

Toronto, Canada

- Dean's List Scholar
- **Relevant Coursework:** Introduction to Machine Learning, Computational Linguistics

Swiss Federal Institute of Technology (ETH Zürich)

September 2023 - August 2024

Exchange Year

Zürich, Switzerland

- EESTEC Generative AI hackathon winner
- Awarded 6000 CHF Exchange Scholarship
- **Relevant Coursework:** Natural Language Processing, Probabilistic Artificial Intelligence, Machine Perception, Large Language Models, Advanced Formal Language Theory, Formal Methods and Functional Programming

Experience

Research Project

June 2024 – September 2024

Supervisors: Tianyu Liu, Prof. Dr. Ryan Cotterell

Rycolab, Institute of Machine Learning, ETH Zürich

- Developed a likelihood-tuning method to **increase the accuracy of open language models by up to 3%** on QA tasks.
- Conducted experiments by implementing the proposed mechanism and testing on the NQ-Open and ELI5 datasets.
- Analyzed experimental results and created synthetic data to generalize problem scenarios.

Projects

Combinatory Categorical Grammar Parser

September 2024 – Present

- Implementing the **first** Combinatory Categorical Grammar Parser supervised by Prof. Dr. Gerald Penn.

Speeding up Earley's Algorithm with A* Search | Python

April 2024 – August 2024

- Proposed and implemented a grid search approach, resulting in a **faster** Earley's algorithm.
- Tested and experimented with the proposed algorithms on Treebank datasets to validate the theory.
- Ensured code correctness by implementing tests and debugging issues.

Word Segmentation and Part-of-Speech Tagging with Transformer | Python, PyTorch

October 2023 – January 2024

- Designed a **new decoder embedding mechanism**, allowing the decoder to better capture subtle linguistic nuances and dependencies between characters and tags.
- Implemented a **transformer** model using **PyTorch** with beam-search for generation.
- Achieved leading-edge results with validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging.

Mental Health Support Bot | Python, BeautifulSoup, FastAPI

October 2023

- Led the development of a simple mental health chatbot web app tailored for ETH Computer Science students, assisting team members with debugging.
- Performed web scraping to obtain data for few-shot learning and prompting in NLP models.
- Implemented a real-time chat feature using FastAPI to host the chat platform and model on a server, allowing multiple simultaneous chats with the bot and recall of previous conversations.

Technical Skills

Programming Languages: Python, C, Java, R, SQL

Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask

Tools/Platforms: Git, Docker, Jupyter, GitHub

Databases: PostgreSQL, psycopg2